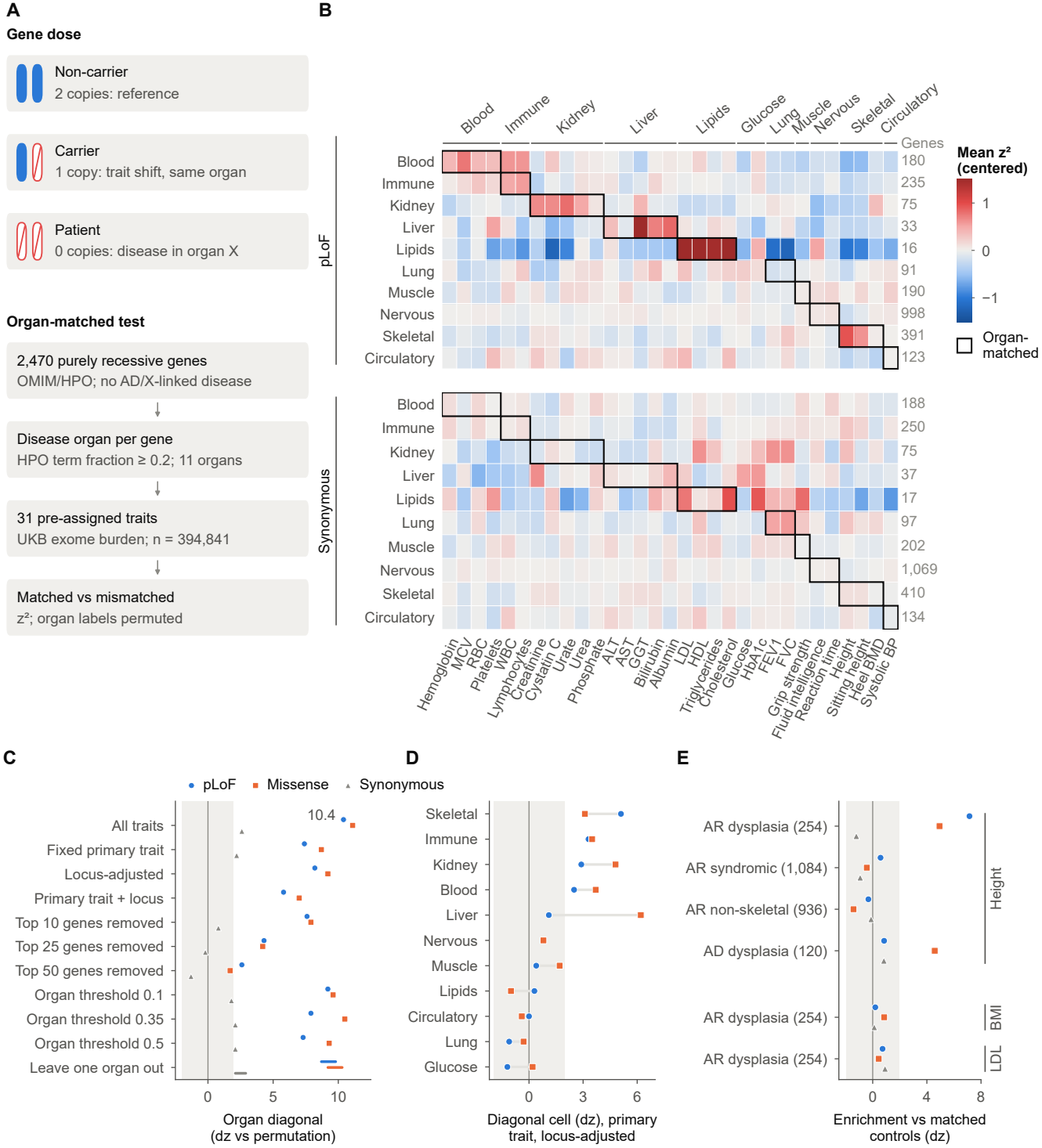

A
Gene dose
Non-carrier
2 copies: reference
Carrier
1 copy: trait shift, same organ
Patient
0 copies: disease in organ X
Organ-matched test
2,470 purely recessive genes
OMIM/HPO; no AD/X-linked disease
Disease organ per gene
HPO term fraction ≥ 0.2; 11 organs
31 pre-assigned traits
UKB exome burden; n = 394,841
Matched vs mismatched
z²; organ labels permuted
B
Blood
Immune
Kidney
Liver
Lipids
Glucose
Lung
Muscle
Nervous
Skeletal
Circulatory
Genes
pLoF
Synonymous
Mean z² (centered)
Organ-matched
Hemoglobin
MCV
RBC
Platelets
WBC
Lymphocytes
Creatinine
Cystatin C
Urate
Urea
Phosphate
ALT
AST
GGT
Bilirubin
Albumin
LDL
HDL
Triglycerides
Cholesterol
Glucose
HbA1c
FEV1
FVC
Grip strength
Fluid intelligence
Reaction time
Height
Sitting height
Heel BMD
Systolic BP
C
pLoF
Missense
Synonymous
All traits
Fixed primary trait
Locus-adjusted
Primary trait + locus
Top 10 genes removed
Top 25 genes removed
Top 50 genes removed
Organ threshold 0.1
Organ threshold 0.35
Organ threshold 0.5
Leave one organ out
10.4
Organ diagonal
(dz vs permutation)
D
Diagonal cell (dz), primary trait, locus-adjusted
E
AR dysplasia (254)
AR syndromic (1,084)
AR non-skeletal (936)
AD dysplasia (120)
AR dysplasia (254)
AR dysplasia (254)
Height
BMI
LDL
Enrichment vs matched controls (dz)